

EE2703: Assignment 10

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1 Introduction

In this assignment, we shall explore the convolution of two discrete-time signals. We shall look at *linear convolution* and *circular convolution* and their relationship. We shall also look at a certain sequence called *Zadoff-Chu sequence*, and its special properties.

2 The FIR Filter

The given FIR Filter is this:

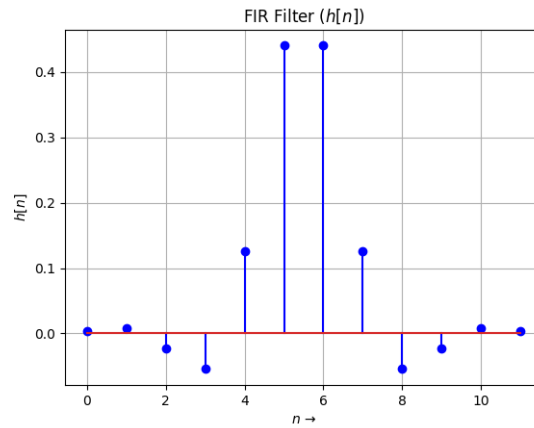


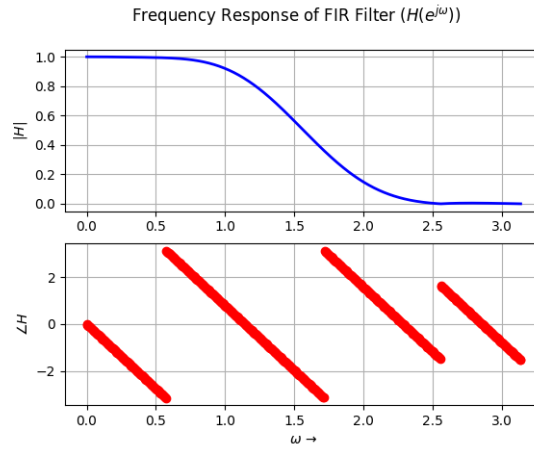
Figure 1: FIR Filter $h[.]$

If we observe carefully, we can see that the envelope of the filter looks like a *sinc*(x) function, which means that its DTFT will look similar to a *rect*(ω).

The magnitude and phase response of the given filter is given by:

$$H(e^{j\omega}) = \sum_{n=0}^{11} h[n]e^{-j\omega n} \quad (1)$$

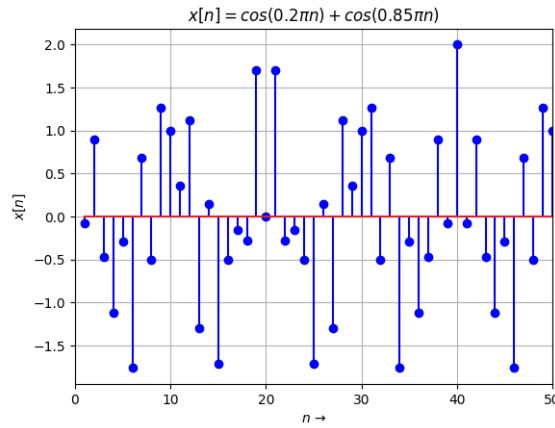
Using the command `scipy.signal.freqz`, we shall be able to get the frequency response as shown below:

Figure 2: Frequency Response of FIR Filter $H(e^{j\omega})$

Since, the sequence $h[\cdot]$ is real, its DTFT $H(e^{j\omega})$ will be even¹. So, we can see that the magnitude response indeed looks like similar to $\text{rect}(\omega)$.

3 Linear Convolution

Now, we shall take an input signal $x[n] = \cos(0.2\pi n) + \cos(0.85\pi n)$, and observe the output of the filter.

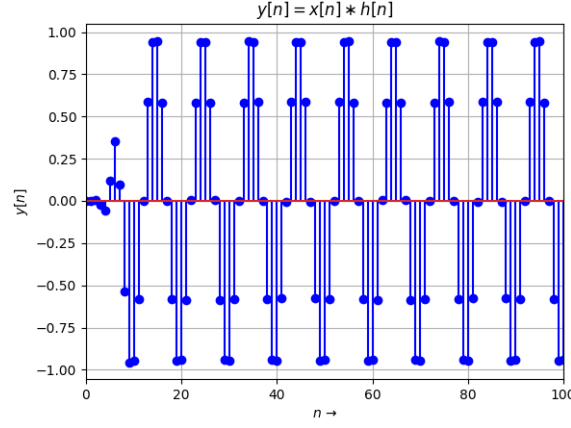
Figure 3: Input to filter, $x[\cdot]$

The output of the filter $y[\cdot]$ can be got by linear convolution of the input $x[\cdot]$ and the filter's impulse response $h[\cdot]$

$$y[n] = x * h = \sum_{\{k: h[k] \neq 0\}} x[n-k]h[k] \quad (2)$$

Using the command `numpy.convolve` to perform the linear convolution operation of $x[\cdot]$ and $h[\cdot]$, we get:

¹ $H(e^{-j\omega}) = \sum_{n=0}^{11} h[n]e^{j\omega n} = H^*(e^{j\omega}) \implies |H(e^{-j\omega})| = |H(e^{j\omega})|$

Figure 4: Output of Filter, $y[n]$

As expected, we see a sinusoid of frequency approximately 0.2π . This is because the filter has damped the high frequency component $\cos(0.85\pi n)$ greatly. We can see from the magnitude response that $|H(e^{j\omega})|_{\omega=0.85\pi}$ is close to 0.

4 Circular Convolution

An N-point circular convolution is defined as:

$$y[\cdot] = x[\cdot] \bigcircled{N} h[\cdot] \quad (3)$$

$$y[n] = \sum_{m=0}^{N-1} x[m]h[(n-m) \bmod N], \quad n \in [0, N-1] \quad (4)$$

Here, the length of all the three signals involved, $x[\cdot]$, $h[\cdot]$ and $y[\cdot]$ are the same and equal to N .

We can easily see that the $y[\cdot]$ obtained as a result of circular convolution and that obtained by linear convolution will not be same always. However, if we take an P-point circular convolution, by appropriately zero padding the two signals, such that $P \geq \text{len}(y_{lin}[\cdot])$, then, the two will match. This is because $y_{cir}[\cdot]$ is the principal period of $\tilde{y}_{lin}[\cdot]$, which is a periodic extension of $y_{lin}[\cdot]$, with period P . So, if the period $P < \text{len}(y_{lin}[\cdot])$, then there will be time-aliasing, which will distort the output. Otherwise, if $P \geq \text{len}(y_{lin}[\cdot])$, then, there will not be any time-aliasing and so, $y_{cir}[\cdot]$ and $y_{lin}[\cdot]$ will have the same information.

We get the following graphs, depending on the length of the circular convolution output.

Case (a) $N = 1024$

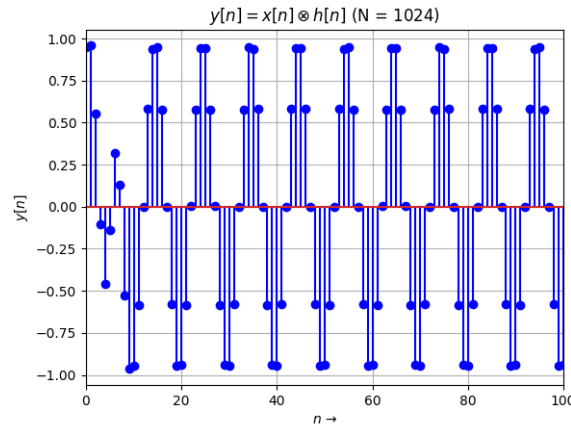


Figure 5: 1024-point circular convolution

Case (b) $N = 1034$

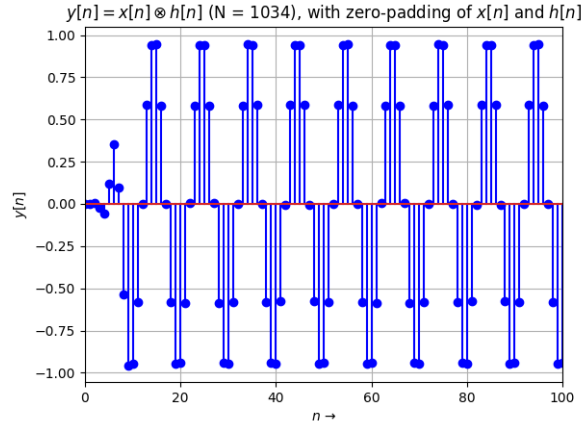


Figure 6: 1034-point circular convolution

Here, we can see that the 1034-point circular convolution results in the same output as linear convolution. This is because, the length of linear convolution signal is $1024 + 11 - 1 = 1034$. Thus, we have implemented linear convolution using circular convolution.

But, to efficiently implement linear convolution through circular convolution, we have to take 2^m -point circular convolution. This is because, the underlying algorithm, the *Fast Fourier Transform* (FFT) is most efficient when it is used for 2^m -point signals.

5 Circular Correlation of Zadoff-Chu Sequence

Consider the Zadoff-Chu sequence, a commonly used sequence in communication. The properties of the sequence are:

1. It is a complex sequence.
2. It is a constant amplitude sequence.
3. The auto correlation of a Zadoff-Chu sequence with a cyclically shifted version of itself is zero, except at the shift.
4. Correlation of Zadoff-Chu sequence with a delayed version of itself will give a peak at that delay.

The Zadoff-Chu sequence is shown below:

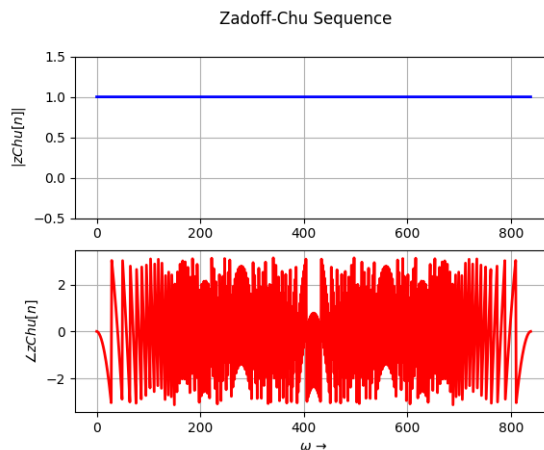


Figure 7: Zadoff-Chu Sequence

We can verify property (3) given above, by plotting the auto-correlation of $zChu[.]$ with a cyclically delayed version of itself. I have taken the auto-correlation with $zChu[.]$ cyclically rotated by 5. The result was:

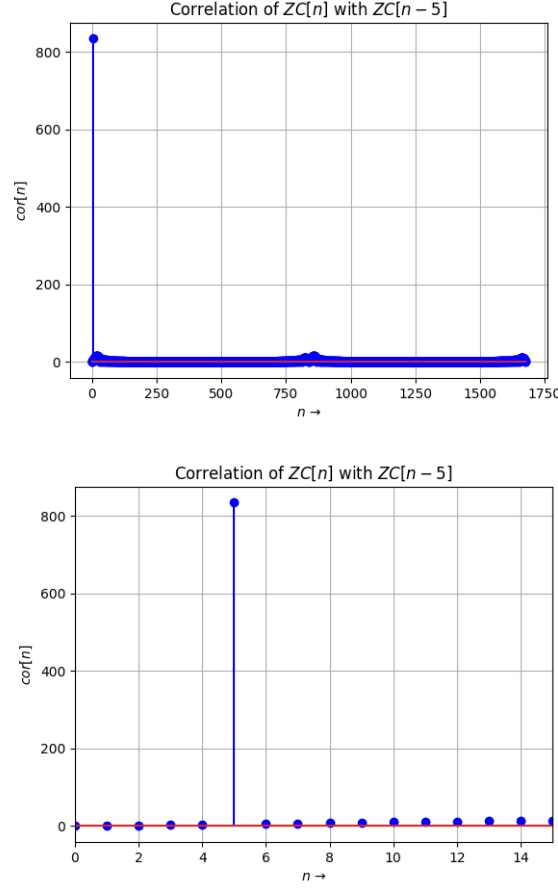


Figure 8: Auto-correlation of $zChu[.]$ with a cyclically delayed version of itself

In the above figure, we can see that the auto-correlation is non-zero only at $n = 5$, which corresponds to the delay value.

6 Conclusion

Linear convolution algorithm implemented using a direct summation is non-optimal and computationally expensive. A faster way to perform the convolution is to use the DFTs of the input and the filter. Circular convolution can be used for the implementation of linear convolution, with a much faster computation speed. The magnitude and phase response of a low pass filter were studied and the system's output, for a mixed frequency signal was obtained through three different methods. For the Zadoff-Chu sequence, its auto-correlation with a cyclically shifted version of itself was found to be non-zero only at the point corresponding to the shift.